

PICOPATT / Picoclimate

Shapelets EDA & Clustering Summary — May 29, 2026

Explainable Clustering for Spatio-Temporal Picoclimatic Data

Project Team

May 29, 2026

Run summary

- Data: `data/picoclimate_test/tracks_measurements.csv` -> aggregated windows
- Rows (windows): 165; Slot-columns: 100
- Selected shapelets: 74; KMeans best $k=2$; KMeans silhouette=0.313
- Ward silhouette=0.313; HDBSCAN: unavailable in env

Data structure and daily windows

- Raw fixture is track-based: each `track_id` is an ordered sequence of measurements.
- If `loc_id` follows the visit order, it can be treated as a pseudo-time slot axis for a track.
- For daily summaries, group by `track_id + date + time_slot` and keep a fixed slot order.
- If the raw order is irregular, align first, then resample to a common length before extracting shapelets.

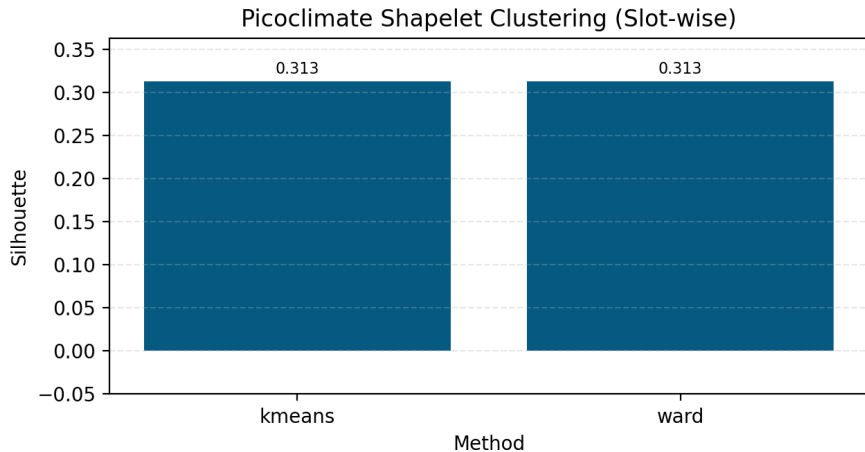
Practical rule

Use the real time columns when available; use `loc_id` ordering only as a fallback pseudo-time index.

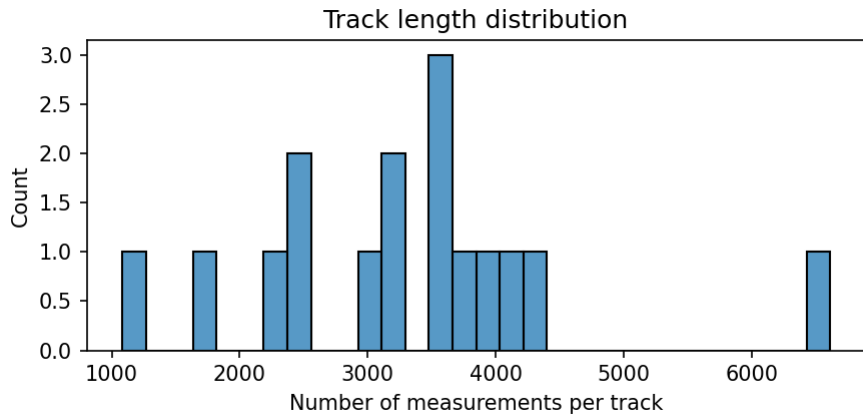
Why `window_features.csv` still matters

- It is a fixed-width table built from the track fixture, so clustering and EDA are reproducible and cheap.
- It lets us compare shapelet-distance clustering against a simpler baseline on the same windows.
- It decouples two questions: representation quality and clustering quality.
- For today's meeting, it is the stable input that feeds the current shapelet pipeline and slide figures.

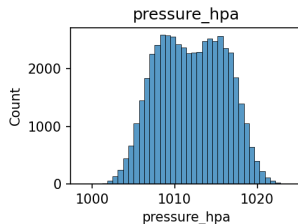
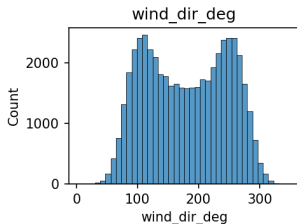
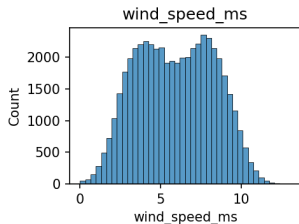
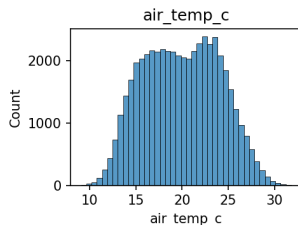
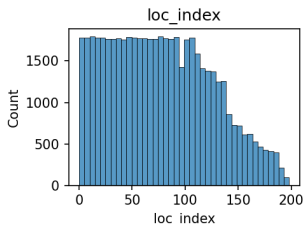
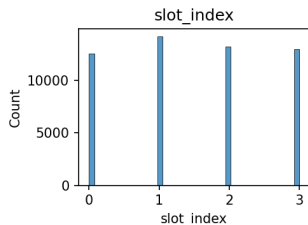
Silhouette by method



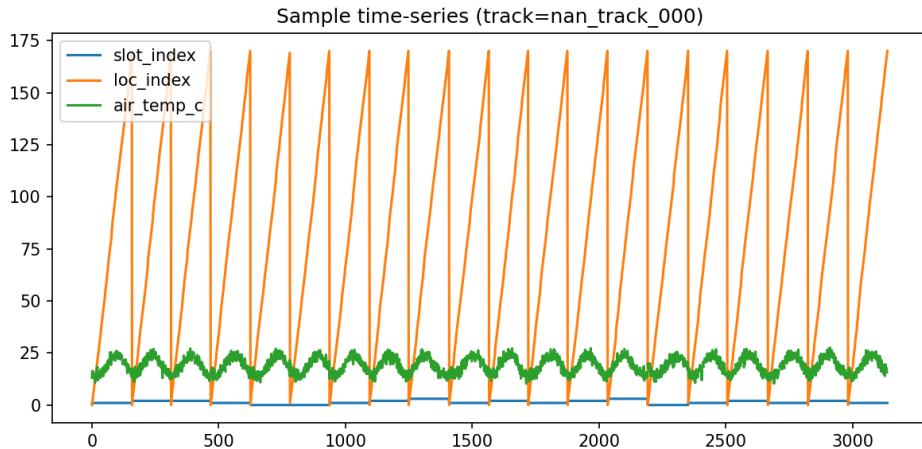
EDA — Track length distribution



EDA — Numeric histograms (top variables)



EDA — Sample time-series



Shapelet run statistics

- Selected shapelets: 74
- Candidate pool: 1000; rows (windows): 165; slot-columns: 100

KMeans silhouette by k (selected)

k	silhouette
2	0.3133
3	0.1493
4	0.1586
5	0.1374
6	0.1280

Detailed outputs: docs/reports/picoclimate₂₀₂₆₋₀₅₋₂₉/

Key approach

- Endpoint extension + Catmull–Rom spline resampling to homogenize mixed-length segments.
- Smooth + resample each indicator to fixed length prior to shapelet scoring and clustering.
- Implementation: see `scripts/alignment.txt` for `rescale_and_smooth` and `rescale_and_smooth_2d` helpers.

Example functions: `extend_points`, `rescale_and_smooth`, `rescale_and_smooth_2d`
(Catmull–Rom)

Next steps for meeting

- Present EDA figures: track-lengths, numeric histograms, sample time-series.
- Explain pipeline: tracks -> window_features -> mixed-length shapelets -> shapelet-distance matrix -> KMeans -> surrogate explanations (SHAP and abductive edits).
- Discuss clustering strategies: by month/year, by location subset, or data-driven (subspace clustering); cover trade-offs.
- Immediate actions: add per-shapelet stats (mean/std/max), export cluster summaries CSV, and prepare brief report slides.